

Tao2.6

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1 Linear Transformations

Let $f : E \rightarrow \mathbb{R}^m$ be a function with $E \subset \mathbb{R}^n$ and $x_0 \in E$ a limit point of E . f is said to be differentiable with derivative $L \in \mathbb{R}^{m \times n}$ if

$$\lim_{x \rightarrow x_0; x \in E - x_0} \frac{\|f(x) - (f(x_0) + L(x - x_0))\|_2}{\|x - x_0\|_2} = 0.$$

Suppose f is diff at x_0 with derivatives L . Then equivalently,

$$\lim_{x \rightarrow x_0; x \in E - x_0} \frac{f(x) - (f(x_0) + L(x - x_0))}{\|x - x_0\|_2} = 0.$$

Let $\epsilon > 0$. Pick $\delta > 0$ such that $0 < \|x - x_0\|_2 < \delta \rightarrow \frac{\|f(x) - (f(x_0) + L(x - x_0))\|_2}{\|x - x_0\|_2} < \epsilon$. Then

$$\left\| \frac{f(x) - (f(x_0) + L(x - x_0))}{\|x - x_0\|_2} \right\| = \frac{\|f(x) - (f(x_0) + L(x - x_0))\|_2}{\|x - x_0\|_2} < \epsilon.$$

Conversely, suppose $\lim_{x \rightarrow x_0; x \in E - x_0} \frac{f(x) - (f(x_0) + L(x - x_0))}{\|x - x_0\|} = 0$. Since $\|\cdot\|_2$ is continuous,

$$\lim_{x \rightarrow x_0; x \in E - x_0} \frac{\|f(x) - (f(x_0) + L(x - x_0))\|_2}{\|x - x_0\|_2} = \left\| \lim_{x \rightarrow x_0; x \in E - x_0} \frac{f(x) - (f(x_0) + L(x - x_0))}{\|x - x_0\|_2} \right\|_2 = \|0\|_2 = 0.$$

Suppose f is differentiable at x_0 with derivatives L, L' . Then

$$\lim_{x \rightarrow x_0; x \in E - x_0} \frac{(L - L')(x - x_0)}{\|x - x_0\|_2} = 0,$$

(derivative is not necessarily unique at boundary points...)

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be given by $f(x, y) = (x^2, y^2)$. I claim that $f'(x_0, y_0) =$

$\begin{bmatrix} 2x_0 & 0 \\ 0 & 2y_0 \end{bmatrix}$. Observe that

$$\begin{aligned} & \lim_{z \rightarrow z_0; z \in \mathbb{R}^2 - z_0} \frac{\|(x^2, y^2) - (x_0^2, y_0^2) - (2x_0(x - x_0), 2y_0(y - y_0))\|_2}{\|(x, y) - (x_0, y_0)\|_2} \\ &= \lim_{z \rightarrow z_0; z \in \mathbb{R}^2 - z_0} \frac{\|((x - x_0)^2, (y - y_0)^2)\|_2}{\|(x - x_0, y - y_0)\|_2} \\ &\leq \lim_{z \rightarrow z_0; z \in \mathbb{R}^2 - z_0} \frac{(x - x_0)^2 + (y - y_0)^2}{\|(x - x_0, y - y_0)\|_2} = \lim_{z \rightarrow z_0; z \in \mathbb{R}^2 - z_0} \|z\|_2 = 0, \end{aligned}$$

so by Squeeze lemma the original limit is zero.

Consider the situation with two derivatives at $x_0 \in E^\circ$. Then $L = L'$. Proof: Sps FTSOC $L \neq L'$, so $(L - L')e_i \neq 0$ for some $i \in [n]$. Let $\epsilon = \|(L - L')e_i\|_2 > 0$. Pick $\delta > 0$ such that $0 < \|x - x_0\|_2 < \delta \rightarrow \frac{\|(L - L')(x - x_0)\|_2}{\|x - x_0\|_2} < \epsilon$ and $V_\delta(x_0) \subset E$. Then for $x - x_0 = (\delta/2)e_i$ (so $x \in E$), we have

$$\frac{\|(L - L')(\delta/2)e_i\|_2}{\|(\delta/2)e_i\|_2} = \|(L - L')e_i\|_2 < \epsilon,$$

absurd.

Suppose $f, g : E \rightarrow \mathbb{R}^m$ with derivatives L, M at $x_0 \in E^\circ$, respectively. Then $f + g : E \rightarrow \mathbb{R}^m$ is differentiable with the derivative $L + M$.

Suppose

$$\lim_{x \rightarrow x_0; x \in E - x_0} \frac{f(x) - (f(x_0) + L(x - x_0))}{\|x - x_0\|} = 0$$

and

$$\lim_{x \rightarrow x_0; x \in E - x_0} \frac{g(x) - (g(x_0) + M(x - x_0))}{\|x - x_0\|} = 0.$$

Then

$$\lim_{x \rightarrow x_0; x \in E - x_0} \frac{(f + g)(x) - ((f + g)(x_0) + (L + M)(x - x_0))}{\|x - x_0\|} = 0,$$

ie $(f + g)'(x_0) = L + M$.

Suppose $f : E_1 \rightarrow E_2, g : E_2 \rightarrow E_3$. Let $x_0 \in E_1^\circ$ and $y_0 = f(x_0) \in E_2^\circ$. Suppose that f, g are differentiable at x_0, y_0 with derivatives L, M , respectively. Then $g \circ f : E_1 \rightarrow E_3$ is differentiable at x_0 with derivative ML .

Let $\epsilon > 0$. Since g is differentiable at y_0 , pick $\delta_1 > 0$ such that $y \in V_{\delta_1}(y_0) - y_0 \rightarrow \|g(y) - (g(y_0) + M(y - y_0))\| \leq \epsilon \|y - y_0\|$. Since f is differentiable at x_0 pick $\delta_2 > 0$ such that $x \in V_{\delta_2}(x_0) \rightarrow \|f(x) - f(x_0)\| \leq \delta_1$.